

Muhammad Hasnain Rafique

Python Backend Developer | Flask & FastAPI | RESTful APIs
hasnainrafique608@gmail.com | +92-321-9378515 | [linkedin.com/in/hasnainrafique](https://www.linkedin.com/in/hasnainrafique) | github.com/hasnainrafique

PROFESSIONAL SUMMARY

Final-year BS Computer Science student (GCUF, 2026) with hands-on experience building and deploying full-stack web applications using Python, Flask, and FastAPI. Delivered 2 production-ready projects featuring RESTful APIs, MySQL databases, and role-based access control. Seeking a junior backend or internship role to apply clean, maintainable backend development skills in a professional environment.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), SQL
Frameworks & Libraries: Flask, FastAPI, Flask-Login
Database Management: MySQL, SQLAlchemy ORM, Relational Database Design, CRUD Operations, Schema Design
Frontend Development: HTML5, CSS3, Bootstrap 5, Responsive Web Design, Cross-Device Compatibility
API Development: RESTful API Design, Pydantic Validation, Swagger UI / OpenAPI 3.0, API Integration, Endpoint Testing
Tools & Practices: RBAC, Authentication & Authorization, MVC Architecture, Git, GitHub, VS Code, Postman

PROJECTS

Nexus Solution — College Management System

2026

Final Year Project · Python · Flask · SQLAlchemy ORM · MySQL · HTML5 · CSS3 · Bootstrap 5 · JavaScript

- Built a multi-portal Flask web application with Role-Based Access Control (RBAC) across 3 user roles (Admin, Faculty, Student), with isolated permission scopes and secure session handling via Flask-Login.
- Designed a normalized MySQL database schema with 8+ interconnected SQLAlchemy ORM models covering student enrollment, course management, timetable scheduling, and announcements.
- Structured the codebase using MVC architecture, separating business logic, SQLAlchemy ORM models, and Jinja2 templates for maintainability and scalability.

Finance Management System — Personal Finance Tracker

2025

Python · FastAPI · REST APIs · MySQL · Pydantic · Swagger UI · HTML5 · CSS3 · Bootstrap 5 · Chart.js

- Developed a RESTful financial platform with 12+ FastAPI endpoints for income tracking, expense categorization, and budget management, using Pydantic models for automatic request validation.
- Integrated Swagger UI / OpenAPI 3.0 documentation, enabling interactive endpoint testing and self-describing API contracts without additional tooling.
- Built an interactive Chart.js analytics dashboard for real-time income-vs-expense tracking, category breakdowns, and budget reporting.

EDUCATION

Bachelor of Science in Computer Science

Government College University Faisalabad (GCUF), Punjab, Pakistan

Oct 2022 – Present

Expected: 2026

Relevant Coursework: Data Structures & Algorithms · Object-Oriented Programming · Database Management Systems · Software Engineering · Operating Systems · Computer Networks · Web Development

CERTIFICATIONS

- Fundamentals of Python Programming freeCodeCamp · 2026
- Introduction to Databases and SQL Great Learning · 2025